Introduction to Group theory, Lecture notes

Università degli Studi di Torino
Academic Year 2026/2027
Prof. Panero, notes by Paolo Rocchietti March[1]

---

[1] paolo.rocchiettimarc@edu.unito.it

Contents

Lesson 1, 23/09 **4**

Lesson 2, 28/09 **4**

1 Again on SU(2) **4**

2 Generators **4**

3 Lorentz-Poincaré Group **4**

4 Internal degrees of freedom **6**

5 Other definitions on groups **6**

6 Examples of groups **7**

Lesson 3, 30/09 **7**

7 Other examples **7**

8 Subgroups **9**

9 Maps between groups **9**

10 Definitions **10**

11 The smallest finite groups **10**
11.1 Cyclic groups 10

Lesson 4, 05/10 **11**
11.2 Symmetric groups 12
11.3 Partitions and Young diagrams 13
11.4 Free Groups 13

Lesson 5, 07/10 **14**
11.5 Braid Groups 14

12 Continuous Groups **15**
12.1 Simplest Lie groups 16
12.2 Matrix Groups 16
12.3 Action of Groups 17

Lesson 6, 19/10 **19**

Lesson 7, 21/10 **20**

Lesson 8, 26/10 **20**

Lesson 9, 28/10 **20**

13 in medias res **20**

Lesson 10, 02/11 **21**
13.1 Characters 22

14 Tensors **26**

Lesson 1, 23/09

ON TABLET NOTES

Lesson 2, 28/09

1 Again on SU(2)

The matrices of SU(2) can be seen as matrices in the form

$$g = \begin{pmatrix} x_0 + ix_3 & x_2 + ix_1 \\ -x_2 + ix_1 & x_0 - ix_3 \end{pmatrix}, \text{ with } x_0^2 + x_1^2 + x_2^2 + x_3^2 = \det(g) = 1$$

It can be also represented by bigger matrices, but some of them are trivial, like

$$g = \begin{pmatrix} x_0 + ix_3 & x_2 + ix_1 & 0 \\ -x_2 + ix_1 & x_0 - ix_3 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \text{ with } x_0^2 + x_1^2 + x_2^2 + x_3^2 = \det(g) = 1$$

which is the same as before, but with a bunch of zeros.

2 Generators

The generators are defined as the "directions" in which matrices can "move away" from the identity matrix: it is clear that

$$g = x_0 \mathbb{I} + i x_i \sigma^i$$

where σ^i are the Pauli matrices; since this is a continuous group, the infinitesimal movement in the direction σ^1 can be written as

$$g = \mathbb{I} + i\varepsilon\sigma^1 + O(\varepsilon^2).$$

Consider the set of generators of a continuous (Lie) group $\{T^a\}$: the commutator between them is

$$[T^a, T^b] = T^a T^b - T^b T^a \underset{Lie}{=} i f^{ab}{}_c T^c$$

in the case of $SU(2)$, the structure constants turn out to be $f^{ab}{}_c = \varepsilon^{ab}{}_c$, which is the Levi-Civita symbol.

The algebra of the generators of $SO(3)$ is exactly the same.

3 Lorentz-Poincaré Group

This is the group of transformations of Special Relativity: using 4-vectors, a kind of transformation that obeys the rules of SR is a boost along the x-axis

$$B_1 = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

this is a kind of rotation in a Minkowski space; regular spacial rotations also obey SR

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R & \\ 0 & & & \end{pmatrix}$$

where $R \in SO(3)$. There are two other transformations, the parity and time-inversion:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \qquad \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

This group has 6 generators: 3 rotations from $SO(3)$ and 3 boosts (along the 3 axes); what are the generators for the boosts?

From their definitions,

$$\gamma^2 - (\gamma\beta)^2 = 1$$

so they can be parametrized like

$$\begin{cases} \gamma = \cosh(\eta) = 1 + O(\eta^2) \\ \beta\gamma = \sinh(\eta) = \eta + O(\eta^2) \end{cases}$$

where η is called rapidity.

$$B_1(\eta) = \mathbb{I} + i\eta \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + O(\eta^2)$$

So, the generators of rotations are $\{J_i\}$, with the aforementioned algebra.

Similarly, the generators of the boosts $\{K_i\}$ have commutation relations

$$[K^a, K^b] = -i\varepsilon^{ab}{}_c J^c$$

Trying to mix the generators yields

$$[J^a, K^b] = i\varepsilon^{ab}{}_c K^c.$$

This can be written in a cleaner way, defining

$$N_a = \frac{J_a + iK_a}{2}$$
$$M_a = \frac{J_a + -iK_a}{2},$$

it can be proven that they satisfy 2 closed $SU(2)$ algebras:

$$[N^a, N^b] = i\varepsilon^{ab}{}_c N^c$$
$$[M^a, M^b] = i\varepsilon^{ab}{}_c M^c$$
$$[N_a, M_b] = 0$$

Since $SU(2)$ is the angular momentum group, an "azimuthal number" can be defined for the two algebras, p, q: depending on the values of these eigenvalues the field satisfying the Lorentz invariance has different properties.

p	q	Name
0	0	Scalar
1/2	0	Spin 1/2 right
0	1/2	Spin 1/2 left

The addition of the Poincaré group are spacetime translations, which are related to the 4-momentum P^μ

4 Internal degrees of freedom

Some particles are invariant under transformations of "internal" degrees of freedom, like colour charge in quarks: if these transformations are local, they are called **gauge transformations**; locally it is always possible to redefine what the colour "blue" is for a quark.

Why is it possible to assume local symmetries? Because it makes the mediator fields come out of the equations (what?).

In the standard model, these internal degrees of freedom are

$$\underbrace{SU(3)}_{\text{QCD}} \times \underbrace{SU(2) \times U(1)}_{\text{Electroweak}}$$

5 Other definitions on groups

Def. A group $(AG4, \cdot)$ is Abelian if and only if

$$\forall g_1, g_2 \in AG4,\ g_1 \cdot g_2 = g_2 \cdot g_1$$

Observation 1 The definition of group implies the uniqueness of the identity element.
Proof (by contradiction) Consider 2 identity elements

$$e \neq f :$$

then their product must be, at the same time

$$\begin{cases} e \cdot f = f \\ e \cdot f = e \end{cases}.$$

This contradicts the assumption $e \neq f$ $\square$

Observation 2 The inverse of every element is unique. The proof is similar to the previous one

Def. A magma is a set with an internal binary operation.
Def. A magma $(M, \cdot)$ is called a semi-group if and only if the binary operation is associative
Def. A semi-group with an identity element is called a monoid.

$$\text{Magma} \supset \text{Semi-group} \supset \text{Monoid} \supset \text{Group} \supset \text{Abelian Group}$$

The set of a group can be

- Finite;
- infinite and discrete;
- infinite and continuous.

The smallest non-trivial group is called $\mathbb{Z}_2$:

$$\mathbb{Z}_2 = \{e, a\}$$

and is called cyclic group of order 2; in order to specify the algebraic structure, all the possible products must be determined: a useful tool is a Caylem table In this case, the only element that has to be explicitly

	e	a
e	$e \cdot e$	$e \cdot a$
a	$a \cdot e$	$a \cdot a$

specified is $a \cdot a$. This must be $a \cdot a = e$, because supposing $a \cdot a = a$ implies

$$\begin{gathered} a \cdot a = a \\ (a \cdot a) \cdot a^{-1} = a \cdot a^{-1} \\ a \cdot (a \cdot a^{-1}) = e \\ a \cdot e = e \\ a = e \end{gathered}$$

which is absurd.

This group can be realized in many ways (see lesson 1) and also with

$$(\{\text{False}, \text{True}\}, \text{XOR})$$

Def. Given two groups $(G_1, \cdot)$, $(G_2, \circ)$, their **direct group product** is the group $G_1 \times G_2$ made from the set of ordered pairs

$$(a_1; a_2), \text{ with } a_i \in G_i,$$

and the internal operation $\square = (\cdot\,; \circ)$ that works as follows

$$(a_1; a_2) \square (b_1; b_2) := (a_1 \cdot b_1; a_2 \circ b_2)$$

6 Examples of groups

Integers: The set $\mathbb{Z}$ is an Abelian group with the regular sum $+$, identity element 0 and inverse $n^{-1} := -n$.

Reals: The set of real numbers is similarly an Abelian group with the regular sum, but to construct a group with the multiplication the number 0 must be taken out: $(\mathbb{R} \setminus 0, \cdot)$ is an Abelian group, with 1 as the identity and $a^{-1} = \frac{1}{a}$.

n × m matrices over $\mathbb{R}$: These sets are groups with the matrix sum, as they are equivalent to $n \times m$ copies of $(\mathbb{R}, +)$.

Invertible matrices: These sets are groups with the matrix multiplication and the inverse is the matrix inverse.

Lesson 3, 30/09

7 Other examples

Dihedral groups: The set is

$$D_n = \{\text{Rotations and reflections that preserve the shape of a regular n-sided polygon}\}$$

and the group product is the "composition" of these functions.
The smallest group of this family, D_3, has 6 elements: 3 rotations of angles 0, $\frac{2\pi}{3}$, $\frac{4\pi}{3}$ and 3 reflections about the axes of the triangle.

In D_4, there are 4 rotations, but the number of reflections is different: 2 about diagonal axes (as the other 2 are the same) and 2 about the axes of the sides (as the other 2 are the same).

In general, $\forall n \geq 3$, the group D_n has $2n$ elements, so its order is

$$|D_n| = 2n :$$

n rotations of angles $\frac{2k\pi}{n}$, with $k = 0, 1...n-1$ and n reflections. No dihedral group is Abelian, since rotations and reflections do not commute.

Linear transformations of orthonormal bases in $\mathbb{R}^2$: There are only two kinds of transformations that preserve the orthonormality

$$M(\alpha) = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & cos\alpha \end{pmatrix} \qquad N(\alpha) = \begin{pmatrix} \cos\alpha & \sin\alpha \\ \sin\alpha & -cos\alpha \end{pmatrix}, \text{ with } 0 \leq \alpha \leq 2\pi$$

the first has $\det(M) = 1$, while the second has $\det(N) = -1$. The group product is again the matrix multiplication.
This group is called the **2-dimensional orthogonal group over $\mathbb{R}$ $\mathbf{O(2}, \mathbb{R})$**, and the matrices with $\det(M) = 1$ form a subgroup called **special orthogonal group**

$$SO(2, \mathbb{R}) = \{M \in O(2, \mathbb{R}) | \det(M) = 1\}$$

The special group is Abelian (it is easy to show using the properties of trigonometric functions), while $O(2, \mathbb{R})$ is not because of the presence of "spatial inversions".

Linear transformations of orthonormal bases in $\mathbb{C}^2$: They can be expressed as matrices in the form

$$M = \begin{pmatrix} a_0 + ia_3 & a_2 + ia_1 \\ -a_2 + ia_1 & a_0 - ia_3 \end{pmatrix}, \text{ with } a_i \in \mathbb{R} \text{ s.t. } a_i a^i = 1$$

note that a_i satisfy the equation of a $3-$sphere in $\mathbb{R}^4$; this can also be expressed as

$$M = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}, \text{ with } |\alpha|^2 + |\beta|^2 = \det(M) = 1.$$

It is easy to show that matrices of this form are unitary:

$$MM^\dagger = M^\dagger M = \mathbb{I}.$$

It was shown before that

$$M = a_0\mathbb{I} + ia_i\sigma^i + O(\mathbf{a}^2) :$$

this is equivalent to

$$M = \exp\left[i\varphi\hat{v}_i\sigma^i\right]$$

where $0 \leq \varphi \leq \pi$ is an angle, $\hat{\mathbf{v}}$ is a unit vector in $\mathbb{R}^3$ and the exponential of a matrix is defined as

$$e^M := \sum_{i=n}^{\infty} \frac{M^n}{n!}.$$

Computing the exponential, it can be shown that these two expressions are equivalent: doing so is, in this case, quite easy using the properties of Pauli matrices

$$\left[\sigma^i, \sigma^j\right] = 2i\varepsilon^{ij}{}_k\sigma^k$$
$$\left\{\sigma^i, \sigma^j\right\} = 2\delta^{ij}\mathbb{I} \rightarrow (\sigma^i)^2 = \mathbb{I}$$

from the second property follows that

$$\sigma^i\sigma^j = \frac{1}{2}\left(\sigma^i\sigma^j - \sigma^j\sigma^i + \sigma^i\sigma^j + \sigma^j\sigma^i\right) = \frac{1}{2}\left(\left[\sigma^i, \sigma^j\right] + \left\{\sigma^i, \sigma^j\right\}\right)$$

From this and some calculations follows that

$$\exp\left[i\varphi\hat{v}_i\sigma^i\right] = \cos\varphi\mathbb{I} + i\sin\varphi\hat{v}_i\sigma^i.$$

Doing so explicitly for $\hat{v} = \begin{pmatrix}1 & 0 & 0\end{pmatrix}$ yields

$$M = \sum_{n=0}^{\infty}\frac{(i\varphi\sigma^i)^n}{n!} = \sum_{k=0}^{\infty}\frac{(i\varphi\sigma^i)^{2k}}{(2k)!} + \sum_{k=0}^{\infty}\frac{(i\varphi\sigma^i)^{2k+1}}{(2k+1)!}$$

since $(\sigma^1)^{2k} = \mathbb{I}$ and $(\sigma^1)^{2k+1} = \sigma^1$, this becomes

$$M = \mathbb{I}\sum_{k=0}^{\infty}\frac{(-1)^k\varphi^{2k}}{(2k)!} + i\sigma^i\sum_{k=0}^{\infty}\frac{(-1)^k\varphi^{2k+1}}{(2k+1)!} = \cos\varphi\mathbb{I} + i\sigma^i\sin\varphi$$

8 Subgroups

Def. Given a group $(G,\cdot)$, a subset $H \subseteq G$ is a *subgroup* of G if and only if $(H,\cdot)$ is a group.

Observation: The trivial subgroups are $H = G$ and $H = \{e\} \subseteq G$; all other subgroups are called *proper* subgroups.

Def. A subgroup $H \subseteq G$ is called a *normal* subgroup if and only if

$$\forall h \in H,\ \forall g \in G,\ g\cdot h\cdot g^{-1} \in H.$$

Observation: If G is Abelian, every subgroup is normal.

Def. A group G is called a *semi-simple* group if and only if it does not admit a proper, normal and Abelian subgroup.

Def. A group G is called a *simple* group if and only if it does not admit a proper and normal subgroup.

Observation: All simple groups are semi-simple groups, but not vice-versa: a group that admits at least one proper, normal, non-Abelian subgroup and no Abelian subgroups is semi-simple but not simple!

9 Maps between groups

Def. A group *homomorphism* f is a map between two groups

$$f : (G,\cdot) \to (Q,\circ)$$
$$f : g \in G \to q \in Q$$

such that

$$\forall g_1, g_2 \in G,\ f(g_1\cdot g_2) = f(g_1)\circ f(g_2).$$

To define a group *isomorphism*, some other definitions are needed.

Def. Given a set $X \neq \emptyset$, a binary relation $\sim$ among its elements is an *equivalence relation* if and only if it is

- Reflexive, i.e. $\forall a \in X,\ a \sim a$;
- Symmetric, i.e. $\forall a, b \in X,\ a \sim b \to b \sim a$;
- Transitive, i.e. $\forall a, b, c \in X,\ a \sim b \wedge b \sim c \to a \sim c$.

Def. Given a set with an equivalence relation $(X, \sim)$, an *equivalence class* is a subset $Y \subseteq X$ such that

$$\forall a, b \in Y \subseteq X,\ a \sim b$$

Observation 1: All sets equipped with an equivalence relation have trivial equivalence classes, which are all the subsets containing only one element.
Observation 2: The equivalence classes of a set X must be disjoint: this follows from the definition of the equivalence relation.

Every equivalence class can be identified by one of its elements, called the *representative* of the class.

Def. A function $f : X \to Y$ is called

- *Injective* if and only if
$$\forall x_1, x_2 \in X,\ f(x_1) = f(x_2) \iff x_1 = x_2;$$
- *Surjective* if and only if
$$\forall \in Y,\ \exists x \in X \text{ s.t. } y = f(x);$$
- *Bijective* if and only if it is injective *and* surjective.

Def. A group *isomorphism* is a bijective group homomorphism.

Def. Two groups are *isomorphic* if $\exists$ an isomorphism f between them.

Theorem The property of isomorphism between two groups is an equivalence relation in the *set of all groups*.
Proof:

- The reflexive property is satisfied by the identity function, which is an isomorphism of a group in itself;
- $\forall G, Q$ groups, if $f : G \to Q$ is an isomorphism, the inverse $f^{-1} : Q \to G$ is an isomorphism;
- $\forall G, Q, L$ groups, if $f : G \to Q$ and $g : Q \to L$ are isomorphisms, then
$$g \circ f : G \underset{f}{\to} Q \underset{g}{\to} L$$
is an isomorphism, since the composition of group homomorphisms is an homomorphism and the composition of bijective functions is bijective. □

10 Definitions

Def. The order of a group G, denoted as $|G|$ is the number of elements in the set of G.

Def. The order of an element $a \in G$, with $a \neq e$, is the smallest $n \in \mathbb{N}$ s.t. $a^n = e$.

11 The smallest finite groups

11.1 Cyclic groups

The smallest finite group is the *trivial group* and has order 1:

$$\mathbb{Z}_1 = \{e\}$$

its Cayley table is trivial

The second smallest group is the group of order 2

$$\mathbb{Z}_2 = \{e, a\}$$

$\cdot$	e
e	e

$\cdot$	e	a
e	e	a
a	a	e

and its Cayley table must be The group of order 3 is

$$\mathbb{Z}_3 = \{e, a, b\}$$

and its Cayley table is

$\cdot$	e	a	b
e	e	a	b
a	a		
b	b		

The missing entries can be found by brute force, i.e. trying all the 3 possible results and seeing if the result contradicts the assumption $a \neq b \neq e$ and/or the properties of the group: there is only one table that is self-consisent and it is

$\cdot$	e	a	b
e	e	a	b
a	a	$a^2 = b$	e
b	b	e	a

For bigger orders, there exist more than 1 self-consistent Cayley table, so how are cyclic groups defined?

Def. The cyclic groups $\mathbb{Z}_n$, with $n = 1, 2...$ are in the form

$$\mathbb{Z}_n = \left\{e, a, a^2, a^3 ... a^{n-1}\right\}$$

where $a^k = \underbrace{a \cdot a \cdot a ....}_{\text{k times}}$.

They can be visualized as the set of the n'th roots of 1 over the complex numbers $\mathbb{C}$:

$$\left\{z_k | z_k = e^{i\frac{2\pi}{n}k}, \text{ with } k = 0, 1...n\right\}$$

with the correspondence $z_0 = e$, $z_k = a^{k-1}$ and with complex multiplication as the group operation.

Lesson 4, 05/10

The order 4 group's Cayley table is not uniquely defined: this means that there exist two order 4 groups, one is $\mathbb{Z}_4$, defined as before stated, and the other is V_4, called the *viergruppe* or the Klein four-group, which is Abelian.

The order 5 group is again uniquely defined and coincides with $\mathbb{Z}_5$; at order 6, along with $\mathbb{Z}_6$, there is the group of symmetries of the equilateral triangle, the dihedral group D_6, which was described earlier: another description is

$$D_6 = \{e, a, b, aba, ab, ba\}$$

with an operation such that

$$a^2 = b^2 = e, \quad (ab)^3 = (ba)^3$$

this underlines the fact that a, b, aba are the reflections, while e, ab, ba are the rotations.

11.2 Symmetric groups

Def. Given a set $X \neq \emptyset$, a *permutation* of the elements of X is a bijective function

$$f : X \to X.$$

The set of permutations of X is denoted as $\mathrm{Perm}(X)$ and it is a group with multiplication defined as function composition.

If $|X| = N \in \mathbb{N}$,

$$|\mathrm{Perm}(X)| = N!$$

and $\mathrm{Perm}(X)$ is also denoted as S_N, *symmetric group* of degree N.

Of course, $S_1 = \mathbb{Z}_1$ and $S_2 = \mathbb{Z}_2$.

A permutation can be represented in two ways:

1. With a $2 \times N$ matrix
$$\begin{pmatrix} 1 & 2 & ... & N \\ x_1 & x_2 & ... & x_N \end{pmatrix}$$
where $x_i = f(i)$.

2. In terms of cycles, for example an element of S_3 can be represented as
$$1 \to 2 \to 3 = (132)$$
which denotes that 1 is mapped to 3, 3 is mapped to 2 and 2 is mapped to 1.

   In matrix representation this is
$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

   Note The cycle notation is *not* guaranteed to contain all elements of the group, for example an element of S_3 can be represented as
$$1 \to 2 = (12) \iff \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}.$$

Observations:

1. Distinct cycles commute
$$(14)(23) = (23)(14)$$

2. Positive permutations of the cycles are equivalent
$$(132) = (321) = (213)$$
and, by convention, they are denoted beginning with the smallest element.

3. Every cycle can be represented as the product of cycles with length 2 by repeating elements:
$$(1324) = (13)(32)(24).$$

This last remark is useful because it can be used to categorize permutations.

Def. The signature of a permutation $P \in \mathrm{S}_N$ is defined as

$$\mathrm{sgn}(P) = (-1)^p, \text{ where } p \text{ is the number of permutations of length 2 that add up to } P.$$

Permutations with $\text{sgn}(P) = 1$ are called *even*, while the others are called *odd*.

It is easy to show that even permutations form a subgroup of S_N, called the *alternating group* A_N, which has order

$$|A_N| = \frac{|S_N|}{2} = \frac{N!}{2}.$$

Cayley's Theorem:
Every finite group of order N is isomorphic to some subgroup of S_N.

Proof. Consider G with $|G| = N$ and the set

$$K = \{f_g : G \to G \text{ s.t. } x \to g \cdot x, \forall g \in G\}$$

which contains N functions f_g: these functions are bijective, because $\forall g \in G$, $(f_g)^{-1} = f_{g^{-1}} \in K$.

For this reason, K is a group with function composition as the group product.

$f_g \in K$ are permutations of G, so K is a subset of $\text{Perm}(G) = S_N$, but, since it is a group with the same operation as S_N, it is also a subgroup of S_N.

Consider the function

$$\mathcal{H} : G \to K \text{ s.t. } g \to f_g$$

this is a group isomorphism: since G is a generic group of order N, the statement is proven. □

Partitions can also be defined for *multigroups*, which are groups in which elements can be repeated: it suffices to use the multinomial coefficient[2]

$$Y = \{1, 2, 2, 3, 3\} \to |Y|(?) = \frac{5!}{1!2!2!}$$

11.3 Partitions and Young diagrams

For every permutation in S_N, the length of the cycles composing it (n_i) always adds up to N

$$\sum_i n_i = N :$$

this means that $n_1, n_2, ... \in \mathbb{Z}^+$ are a *partition* of N.

How many of these partitions are there for a fixed $N \in \mathbb{Z}^+$? The number of partitions is denoted as $p(N)$ and it can be shown that

$$\sum_{N=0}^{\infty} p(N)x^N = \prod_{k=1}^{\infty} \frac{1}{1-x^k}.$$

Each partition of N can be associated to a *Young diagram*: they are tables composed by N boxes, in which the rows have non-increasing length[3].

11.4 Free Groups

Def. Given a group G and a subset

$$X = \{x_1, x_2, ..., x_n\} \subset G,$$

if every element $g \in G \setminus \{e\}$ can be *uniquely* written as a product of the elements of X in the form

$$g = x_{j_1}^{i_1} x_{j_2}^{i_2} ... x_{j_m}^{i_m}$$

[2] Idk why he said this
[3] See section (2.6.2) of the book for more information

in which $i_k \in \mathbb{Z} \setminus \{0\}$ and consecutive elements are different, i.e. $x_{j_k} \neq x_{j_{k+1}}$, $\forall k = 1, 2..., m$, then G is called a *free group* and X is called a *free set of generators of* G.

The elements of X are called *letters* and their product is called a *word*: if the word obeys the constraints earlier stated, it is called a *reduced* word.

Observation The *inverse* of a reduced word, i.e. the inverse element g^{-1}, can be written as

$$g^{-1} = x_{j_1}^{-i_1} x_{j_2}^{-i_2} ... x_{j_m}^{-i_m}$$

Def. The reduced words which give the identity element of a free group are called *relations* r for the group:

$$x_{j_1}^{i_1} ... x_{j_m}^{i_m} = e \to r = e.$$

Def. The *presentation* of a free group is a way of uniquely determine it, using its generators and its independent relations[4]:

$$G = \langle x_1, x_2, ..., x_n | r_1, r_2, ..., r_m \rangle .$$

Example 1 $\mathbb{Z}_N$ is a free group, because the set

$$X = \{a\}$$

generates the whole group: the relation of the group is $a^N = e$, so its presentation is

$$\mathbb{Z}_N = \left\langle a \middle| a^N \right\rangle .$$

Example 2 D_4 can be generated by just 2 letters r, f, which are, respectively, the rotation of $\pi/2$ and the reflection along the vertical axis:

$$D_4 = \left\langle r, f \middle| r^4, f^2, rfrf \right\rangle$$

Example 3 The Pauli group

$$G_1 = \{\pm \mathbb{I}, \pm i\mathbb{I}, \pm \sigma_{1,2,3}, \pm i\sigma_{1,2,3}\} ,$$

relevant for $1-$qubit information theory, has presentation

$$G_1 = \left\langle \sigma_1, \sigma_2, \sigma_3 \middle| \sigma_a^2 = \mathbb{I}, \sigma_a \sigma_b = i\varepsilon_{abc}\sigma_c \, \forall a \neq b) \right\rangle$$ [5]

Lesson 5, 07/10

11.5 Braid Groups

Consider a set of n parallel, vertical strings: some operations can be done to them, like undoing a passover of a string over another (Reidemeister move of the second type) or putting the i'th string over the $i+1$'th, denoted by σ_i.

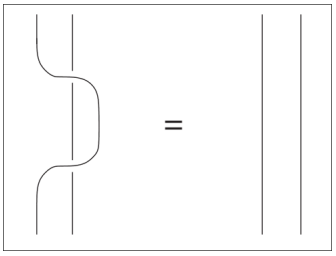


Fig. 11.1: The Reidemeister move of the second type.

[4] Since the relations are *reduced* words, two relations are independent if and only if they are different.
[5] Unclear if relations must always equate to the identity or not.

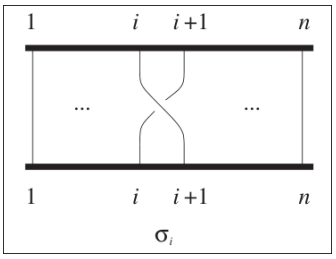


Fig. 11.2: The σ_1 operation.

A generic braid is a word

$$\sigma = \sigma_{j_1}^{i_1} ... \sigma_{j_m}^{i_m}$$

with the generators $X_{n-1} = \{\sigma_1, \sigma_2, ..., \sigma_{n-1}\}$.

To have meaningful braids, the strings must at least be 2: a braid group with n strings is denoted as B_n; the group B_2 is non-trivial:

$$B_2 \cong (\mathbb{Z}, +)$$

Lemma In the group B_n with $n > 2$,

$$\sigma_i \sigma_j = \sigma_j \sigma_i$$

if and only if $|i - j| \geq 2$.

This lemma is intuitively true, since the moves σ_i, σ_j act on different couples of strings if and only if $|i-j| \geq 2$.

Theorem In the group B_n with $n > 2$,

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \forall i$$

This can be proven by executing the moves manually.

With these properties, the presentation of the group can be determined to be

$$B_n = \langle \sigma_1, \sigma_2, ..., \sigma_{n-1} | \sigma_i \sigma_j = \sigma_j \sigma_i \, (|i - j| \geq 2), \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \, (\forall i) \rangle$$

12 Continuous Groups

Def. A group G is said to be continuous if G is infinite and uncountable.

Def. A group G is said to be discrete and continuous if G is infinite and countable, i.e. there exist a one-to-one correspondence with the natural numbers $\mathbb{N}$.

Continuous groups are not necessarily vector spaces, so to characterize their size the term "dimension" is a slight abuse of notation:

Def. Given a continuous group G, its dimension $\dim(G) \in \mathbb{N}$ is the number of independent real parameters, called *coordinates* needed to parametrize its elements.

If the product of two elements of the group

$$g_3 = g_1 \cdot g_2$$

depends *smoothly* on the coordinates of the two elements, the set of coordinates form a differential manifold, called the *group manifold*.

Def. A group G with coordinates that form a group manifold is called a **Lie Group**.

12.1 Simplest Lie groups

The simplest Lie groups have dimension 1, like

$$(\mathbb{R}, +)$$

in which the coordinate choice is trivial, and

$$U(1) := \{z \in \mathbb{C} \text{ s.t. } |z| = 1\}$$

in which the coordinate is $\alpha = \arg(z)$.

These can be combined, respectively, in

$$(\mathbb{R}, +) \times (\mathbb{R}, +) \times ... = (\mathbb{R}^n, +)$$

and

$$[U(1)]^n = U(1) \times U(1)... = \{(z_1, z_2, ..., z_n) \text{ s.t. } |z_i| = 1 \, \forall i\}$$

which is isomorphic to the $n-$torus.

12.2 Matrix Groups

An important class of Lie groups are matrix groups: besides the somewhat trivial group

$$(\text{Mat}(n \times m, \mathbb{R}), +)$$

the interesting groups have matrix multiplication as the group product and for this reason all these groups are sets of invertible $n \times n$ matrices.

1. General linear transformations on $\mathbb{R}^n$

   $$GL(n, \mathbb{R}) := \{A \in \text{Mat}(n \times n, \mathbb{R}) \text{ s.t. } \det(A) \neq 0\} :$$

   this group has dimension n^2, since the inequality $\det(A) \neq 0$ does not reduce the number of independent parameters; as stated before, all matrix groups with multiplication as group product are subgroups of $GL(n, \mathbb{R})$.

2. Special linear transformations on $\mathbb{R}^n$

   $$SL(n, \mathbb{R}) := \{A \in \text{Mat}(n \times n) \text{ s.t. } \det(A) = 1\} :$$

   this group has dimension $\dim(SL(n, \mathbb{R})) = n^2 - 1$, since its dimension as a vector space coincides with the dimension of the manifold.

3. Orthogonal group

   $$O(n) := \left\{A \in \text{Mat}(n \times n, \mathbb{R}) \text{ s.t. } A^T = A^{-1}\right\} :$$

   these transformation preserve the norm of vectors in $\mathbb{R}^n$ and the orthogonality condition implies that $\dim(O(n)) = n(n-1)/2$ and that $\det(A) = \pm 1$; the subgroup with positive determinant is a subgroup of both $SL(n, \mathbb{R})$ and $O(n)$ and it is called the *special* orthogonal group.

4. Special Orthogonal Group

   $$SO(n) := \{A \in O(n) \text{ s.t. } \det(A) = 1\} :$$

   this group has the same dimension of $O(n)$.

5. Unitary group

   $$U(n) := \left\{A \in \text{Mat}(n \times n, \mathbb{C}) \text{ s.t. } A^\dagger = A^{-1}\right\} :$$

   this is the complex equivalent of $O(n)$ and, similarly, preserves the complex norm of $\mathbb{C}^n$ and admits a subgroup with $\det(A) = 1$; it has dimension $\dim(U(n)) = 2n^2 - 2n(n-1)/2 - n = n^2$.

6. Special Unitary Group
$$SU(n) := \{A \in U(n) \text{ s.t. } \det(A) = 1\} :$$
this group has the same dimension of $U(n)$ and it is one of the most important groups for physics, since $SU(2)$ is pivotal for the description of spin and of the Lorentz group; it has dimension $\dim(SU(n)) = n^2 - 1$, since the condition $\det(A) = 1 = e^{i\alpha}$ constrains a parameter.

7. Symplectic group
$$Sp(2n, \mathbb{R}) := \left\{A \in \text{Mat}(2n \times 2n, \mathbb{R}) \text{ s.t. } A^T \Omega A = \Omega, \forall \Omega \text{ symplectic form}\right\} :$$
Ω is the symplectic form that should be familiar from Hamiltonian mechanics
$$\Omega = \begin{pmatrix} \mathbb{O}_n & \mathbb{I}_n \\ -\mathbb{I}_n & \mathbb{O}_n \end{pmatrix}$$
Note For $n = 1$, $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$

There are "exceptional" Lie groups that were discovered after classifying all the Lie Algebras of continuous groups: G_2, F_4, E_6, E_7, E_8; they have some applications in String Theory.

12.3 Action of Groups

Def. Given a group G and a non-empty set X, a *left action* L of G on X is the group homomorphism
$$\begin{aligned} L : G &\to \text{Perm}(X) \\ g &\to L_g. \end{aligned}$$

If L exists, X is called a left G-space.

Since L is a group homomorphism, it follows that
$$L_{g_1 \cdot g_2} = L_{g_1} \circ L_{g_2}$$

Def. Given a group G and a non-empty set X, a *right action* R of G on X is a function
$$\begin{aligned} R : G &\to \text{Perm}(X) \\ g &\to R_g \end{aligned} \quad \text{so that } R_{g_1 \cdot g_2} = R_{g_2} \circ R_{g_1}.$$

If R exists, X is called a right G-space.

Note It can be shown that, if both L, R exist, they are isomorphic[6].

Def. Given a left G-group X, an *orbit* of an element $x \in X$ is the set
$$O_x := \left\{x' \in X \text{ s.t. } \exists g \in G \text{ s.t. } x' = L_g(x)\right\}.$$

Example Given $x \in X = \mathbb{R}^2$ and $G = SO(2)$, its orbit is the set
$$O_x = \left\{x' \in \mathbb{R}^2 \text{ s.t. } |x'| = |x|\right\}$$
since for all points $x' \in \mathbb{R}^2 = X$ that have the same norm as x there exist a rotation, i.e. an element of $SO(2) = G$, that transforms x into x'.

[6]?? How can there be an isomorphism between functions ??

Theorem Belonging to the same orbit is an equivalence relation on X:

1. $x \in O_x$ for all $x \in X$ (reflexive);
2. $x' \in O_x \iff x \in O_{x'}$ (symmetric);
3. $x' \in O_x \wedge x'' \in O_{x'} \to x'' \in O_x$ (transitive).

Proof The first statement can be proven considering that $L_e = \mathrm{Id} \in \mathrm{Perm}(X)$, thus $x \in O_x$.

The second statement can be proven considering $x' \in O_x$ implies $\exists g$ so that $x' = L_g(x)$: the inverse of this action is $(L_g)^{-1} = L_{g^{-1}}$ and

$$(L_g)^{-1}(x') = (L_{g^{-1}} \circ L_g)(x) = x$$

thus $g^{-1} \in G$ is so that $x = L_{g^{-1}}(x')$ and so $x \in O_{x'}$.

The third statement can be similarly proven using the group product: $x' = L_{g_1}(x)$, $x'' = L_{g_2}(x')$, so

$$x'' = (L_{g_2})^{-1}(x') = (L_{g_2^{-1}} \circ L_{g_1^{-1}})(x) = L_{g_2^{-1} \cdot g_1^{-1}}(x) :$$

this means that $g_2^{-1} \cdot g_1^{-1} = g' \in G$ is so that $L_{g'}(x'') = x$, which means $x'' \in O_x$. □

Def. Given a left action L, it is a *transitive left action* if

$$\exists y \in X \text{ s.t. } x \in O_y \, \forall x \in X$$

or, in other words, the set X consists of only one equivalence class with respect to the equivalence relation defined above.

Example Consider $X = \mathbb{R}^2$, $G = (\mathbb{R}^2, +)$ and L as translations on the plane: the left action is transitive, since every point $x \in \mathbb{R}^2$ can be "reached" from any other point with a translation.

Def. The *isotropy group* (or little group) of $x \in X$ is the subset of G whose left actions leave x invariant

$$I_x := \{g \in G \text{ s.t. } L_g(x) = x\}$$

Note I_x is automatically a *subgroup* of G: let $g, g' \in I_x \subseteq G$, so

$$L_g(x) = L_{g'}(x) = x$$

but this also means that

$$L_{g \cdot g'} = L_g(L_{g'}(x)) = L_g(x) = x \implies g \cdot g' \in I_x$$

Furthermore it follows from the definition of left action that

$$L_e(x) = E(x) = x \implies e \in I_x$$

and finally

$$E(x) = (L_g)^{-1}(L_g(x)) = L_{g^{-1}}(x) = x \implies g^{-1} \in I_x$$

Def. If x is so that

$$I_x = G$$

x is called *a fixed point under the action L.*
Example Consider the action of $SO(2)$ on $\mathbb{R}^2$: the origin $(0, 0)$ is a fixed point, because it is invariant under *all* rotations.

12.3.1 Special cases

If $X = G$, there are two special left actions that are always defined:

- Translation: the homomorphism $L : G \to \mathrm{Perm}(G)$ so that $L_g(x) = g \cdot x$;
- Conjugation: the homomorphism $L : G \to \mathrm{Perm}(G)$ so that $L_g(x) = g \cdot x \cdot g^{-1}$

Lesson 6, 19/10

Note Translations are always transitive, i.e. there is only one orbit which consists of the entire group, since

$$g \cdot x = y \implies g = y \cdot x^{-1}$$

or, equivalently,

$$\forall x, y \in G, \exists g \in G \text{ s.t. } L_g(x) = y$$

Proof Conjugation is a left action.
Consider $L_{g \cdot g'}$:

$$L_{g \cdot g'}(x) = (g \cdot g')x(g \cdot g')^{-1} = g \cdot (g' \cdot x \cdot (g')^{-1}) \cdot g^{-1} = L_g(L_{g'}(x)) = L_g \circ L_{g'}(x) \quad \square$$

Conjugation is transitive if G is Abelian, but not in general.

Def. Given $g \in G$, the translation action L and a proper subgroup $H \subset G$, the set

$$gH := \left\{ g' \in G \text{ s.t. } \exists h \in H \text{ s.t. } g' = L_g(h) = g \cdot h \right\}$$

is called a *left coset of* H.

If H is a proper subgroup, there exist more than one left coset. The set of all left cosets is called the *quotient space*:

$$G/H = \{gH, \forall g \in G\}$$

Note This terminology is akin to that of equivalence relations, because the equivalence relation "g is in relation to g' if, for some $h \in H$, $g' = g \cdot h$" forms equivalence classes gH (with representative g) on G, which are disjoint and cover all of G.

Proposition *The elements of gH are in one-to-one correspondence with those of H, thus $|H| = |gH|$.*
Proof Consider $g \in G$ and the function

$$\begin{aligned} f_g :& H \to gH \\ & h \to g \cdot h : \end{aligned}$$

f_g is injective, since

$$f_g(h_1) = f_g(h_2) \implies g \cdot h_1 = g \cdot h_2 \iff h_1 = h_2$$

and surjective (by definition of gH): so f_g is a one-to-one correspondence. $\square$

Lagrange theorem *The order $|H|$ of any proper subgroup $H \subset G$ must be a divisor of $|G|$.*
Proof Since gH are equivalence classes of G, they are disjoint, so

$$|G| = \sum_g |gH| :$$

for the previous proposition, $\forall g, g' \in G$, $|gH| = |H| = |g'H|$ and so

$$|G| = n|gH| = n|H|, \text{ with } n \in \mathbb{N}_0 \quad \square$$

Lesson 7, 21/10

Lesson 8, 26/10

Lesson 9, 28/10

13 in medias res

Last theorem stated that a reducible unitary representation of a group is completely reducible.

Theorem *Let G be a finite group, D a finite-dimensional representation of G: there always exist a scalar product over V such that D is unitary with respect to it.*
Proof Consider $v, w \in V$ and a basis $\{e_i\}$ for V

$$|v\rangle = \sum_{i=1}^{n} v_i \, |e_i\rangle \, , \; |w\rangle = \sum_{i=1}^{n} w_i \, |e_i\rangle \, :$$

a scalar product between v, w is

$$\langle v|w\rangle = \sum_{i=1}^{n} (v^i)^* w^i .$$

Consider the scalar product $\langle\langle v|||w\rangle\rangle$ defined as follows

$$\langle\langle v|||w\rangle\rangle := \frac{1}{|G|} \sum_{g' \in G} \big\langle D(g')v \big| D(g')w \big\rangle \, :$$

consider now

$$\langle\langle D(g)v|||D(g)w\rangle\rangle = \Big\langle \langle v| \Big|\Big| D^\dagger(g) D(g) w \Big\rangle \Big\rangle$$

by the properties of the scalar product. Using the definition of the second scalar product yields

$$\langle\langle D(g)v|||D(g)w\rangle\rangle = \frac{1}{|G|} \sum_{g' \in G} = \big\langle D(g')D(g)v \big| D(g')D(g)w \big\rangle = \frac{1}{|G|} \sum_{g' \in G} \big\langle D(g' \cdot g)v \big| D(g' \cdot g)w \big\rangle$$

since g is fixed and arbitrary, letting $g \cdot g' = g''$ and summing over it does not change anything, so

$$\langle\langle D(g)v|||D(g)w\rangle\rangle = \frac{1}{|G|} \sum_{g'' \in G} \big\langle D(g'')v \big| D(g'')w \big\rangle = \langle\langle v|||w\rangle\rangle$$

so D is unitary with respect to this scalar product. □

Maschke's theorem *Every finite dimensional reducible representation of a finite group is completely reducible.*
Proof Obvious combining the last two theorems.

Schur's Lemma *Given two irreducible representations of D_1, D_2 of a group G and an intertwiner A, either A is an isomorphism or $A = \mathbb{O}$.*
Proof Consider $v \in \ker(A)$:

$$AD_1(g)v = D_2(g)Av$$

since $Av = 0$ by definition of kernel,

$$AD_1(g)v = 0 \iff D_1(g)v \in \ker(A)$$

since v is a generic vector and g is a generic element of G, $\ker(A)$ is a submodule of D_1 ($\forall w \in \ker(A), \forall g \in GD_1(g)w \in \ker(A)$). But D_1 is irreducible, so it admits only V_1 or $\{0\} \subset V_1$ as submodules, so there are only two possibilities for $\ker(A)$:

- $\ker(A) = V_1$, which means that A is the 0-map, $A = \mathbb{O}$;
- $\ker(A) = \{0\}$, so A is injective.

Considering now the case in which A is injective, let $w \in \mathrm{Im}(A)$, then $\exists u \in V_1$ so that $Au = w$, so

$$D_2(g)w = D_2(g)Au = AD_1(g)u :$$

this means that $\exists z = D_1(g)u \in V_1$ so that $D_2(g)w = Az$, thus

$$\forall g \in G, \forall w \in V_2,\ D_2(g)w \in \mathrm{Im}(A)$$

so $\mathrm{Im}(A)$ is a submodule of the irreducible representation of D_2 and, as before, there are only two cases:

- $\mathrm{Im}(A) = V_2$, which means that A is surjective **and** injective;
- $\mathrm{Im}(A) = \{0\}$, which means that A is the 0-map $\mathbb{O}$. □

So A must either be the 0-map or an isomorphism between V_1 and V_2. □

Corollary *Given an irrep (irreducible representation) D, the only operators that commute with $D(g)$, $\forall g \in G$ are proportional to the identity.*
Proof (sketch)
It is clear that $\forall \mu \in \mathbb{C}$

$$\mu\mathbb{I}D(g) = D(g)\mu\mathbb{I};$$

if $AD(g) = D(g)A$, then it is also true that

$$(A - \mu\mathbb{I})D(g) = D(g)(A - \mu\mathbb{I})$$

so by the Lemma either $A = \mu\mathbb{I}$ or $(A - \mu\mathbb{I})$ is an isomorphism: the latter condition is not guaranteed to be satisfied $\forall \mu$, since $\det(A - \mu\mathbb{I}) = 0$ is a polynomial equation in μ of degree $\dim(A)$, that must admit at least one solution for the fundamental theorem of Algebra.
So $A = \mu\mathbb{I}$ for some $\mu \in \mathbb{C}$. □

Fundamental orthogonality theorem *Consider a group G and two irreps U_α, U_β that are not isomorphic*[7] *and act respectively on V_α, V_β: the following equality holds*

$$\sum_{g \in G} (\langle w|U_\alpha(g)v\rangle)^* \langle u|U_\beta(g)t\rangle = \frac{|G|}{\dim(V_\alpha)} \delta_{\alpha\beta} (\langle w|t\rangle)^* \langle v|w\rangle$$

Burnside theorem *Given a finite group G and the set of inequivalent irreps $\{U_\alpha\}$ acting on V_α, the order of the group is*

$$|G| = \sum_\alpha (\dim(V_\alpha))^2$$

Lesson 10, 02/11

[7] Which means that $U_\alpha(g)$ and $U_\beta(g)$ should be so that $\nexists A$ s.t. $\forall g \in G$, $AU_\alpha(g)A^{-1} = U_\beta(g)$

13.1 Characters

Def. The *character* of a representation U_α of a finite group G is the map

$$\begin{aligned}\chi_\alpha :& G \to \mathbb{C} \\ & g \to \chi_\alpha(g) := \mathrm{tr}(U_\alpha(g))\end{aligned}$$

Observation 1 If two representations are linked by a similarity transfomation

$$D_\alpha(g) = AD_\beta(g)A^{-1}$$

their characters are the same, for the properties of the trace

$$\chi_\alpha(g) = \chi_\beta(g)$$

Observation 2 If $g_1, g_2 \in G$ are so that

$$\exists g \in G \text{ s.t. } g_2 = g \cdot g_1 \cdot g^{-1}$$

then

$$\chi_\alpha(g_1) = \chi_\alpha(g_2),\ \forall \alpha$$

and g_1, g_2 are in the same *conjugacy class*, which is an equivalence class.

Observation 3 The character of any representation for any group evaluated at the identity yields

$$\chi_\alpha(e) = \dim(D_\alpha),\ \forall G,\ \forall D_\alpha$$

Observation 4 From the orthogonality theorem follows that

$$\sum_{g\in G} \chi^*_\alpha(g)\chi_\beta(g) = |G|\delta_{\alpha\beta}$$

Observation 5 For a completely reducible representation, i.e.[8]

$$U = \bigoplus_\alpha n_\alpha U_\alpha$$

the linearity of the trace implies

$$\mathrm{tr}\, U(g) = \sum_\beta n_\beta \,\mathrm{tr}(U_\beta(g)) \implies \chi(g) = \sum_\beta n_\beta \chi_\beta(g)$$

multiplying by $\chi^*_a(g)$ and summing over $g \in G$ yields

$$\sum_{g\in G} \chi^*_\alpha(g)\chi(g) = \sum_\beta n_\beta \sum_{g\in G} \chi^*_\alpha(g)\chi_\beta(g) = \sum_\beta n_\beta |G|\delta_{\alpha\beta} = n_\alpha |G|$$

thus

$$n_\alpha = \frac{1}{|G|}\sum_{g\in G} \chi^*_\alpha \chi(g)$$

By considering the conjugacy equivalence classes c_i, $i = 1, 2, ...k$, this becomes

$$n_\alpha = \frac{1}{|G|}\sum_{i=1}^{k} |c_i| \chi^*_\alpha(c_i)\chi(c_i).$$

[8] unclear, maybe it is explained in lessons 6,7,8

By defining the *character vector*

$$\vec{\chi}_\alpha = \begin{pmatrix} \sqrt{|c_1|}\chi_\alpha(c_1) \\ \dots \\ \sqrt{|c_k|}\chi_\alpha(c_k) \end{pmatrix} \in \mathbb{C}^k$$

one can interpret the orthogonality condition literally: this implies that for any finite group **the number of irreps is at most (actually is exactly) the number of conjugacy classes.**

Theorem *If G is Abelian and finite, all its irreps have dimension 1*
Proof If G is Abelian, then all conjugacy classes must contain a single element:

$$g_1 \sim g_2 \iff g_1 = g \cdot g_2 \cdot g^{-1} = g \cdot g^{-1} \cdot g_2 = g_2$$

So $|c_i| = 1$ and

$$|G| = \sum_{i=1}^{k} |c_i| = k$$

so the number of irreps is $|G|$, but, by the Burnside theorem

$$|G| = \sum_\alpha (\dim(V_\alpha))^2$$

since $|G| =$ number of irreps $= \sum_\alpha 1$,

$$\sum_\alpha (\dim(V_\alpha)) = \sum_\alpha 1 \iff \dim(V_\alpha) = 1, \, \forall \alpha$$

□

Note This also holds for infinite groups.

Example: S_3

By Burnside's theorem, the irreps of S_3 can have dimensions

$$1, 1, 1, 1, 1, 1 \text{ or } 1, 1, 2.$$

In S_3 permutations are of three kinds:

1. One permutation with no non-trivial cycles;
2. Three permutations that are cycles of length 2;
3. Two permutations that change every element.

It can be proven that these correspond to the conjugacy classes, so there exist only 3 irreps, which, by Burnside, have dimensions $1, 1, 2$.

The characters of these 3 representations are

	$\chi(E)$	$\chi(1 \to 2)$	$\chi(1 \to 2 \to 3)$
1	1	1	1
$1'$	1	a	b
2	1	c	d

1 is the trivial 1−dimension representation ($\forall g \in G, U(g) = 1 \in \mathbb{R}$), while $1'$ maps all elements to their parity (in the sense of permutations).

Some properties of a, b, c, d can be deduced from the character vectors:

$$\vec{\chi}_1 = \begin{pmatrix} 1 \\ \sqrt{3} \\ \sqrt{2} \end{pmatrix}, \ \vec{\chi}_{1'} = \begin{pmatrix} 1 \\ \sqrt{3}a \\ \sqrt{2}b \end{pmatrix}, \ \vec{\chi}_2 = \begin{pmatrix} 1 \\ \sqrt{3}c \\ \sqrt{2}d \end{pmatrix}$$

since they are orthogonal, it follows that

$$\begin{cases} \vec{\chi}_1^\dagger \cdot \vec{\chi}_{1'} = 1 + 3a + 2b = 0 \\ \vec{\chi}_1^\dagger \cdot \vec{\chi}_2 = 2 + 3c + 2d = 0 \\ \vec{\chi}_{1'}^\dagger \cdot \vec{\chi}_2 = 1 + 3ac + 2bd = 0 \\ \vec{\chi}_1 \cdot \vec{\chi}_1 \equiv 6 \\ \vec{\chi}_{1'} \cdot \vec{\chi}_{1'} = 1 + 3a^2 + 2b^2 = 6 \\ \vec{\chi}_2 \cdot \vec{\chi}_2 = 4 + 3c^2 + 2d^2 = 6 \end{cases}$$

Since the $1'$ representation maps the elements to their parity, $a = -1$ and $b = 1$, so the system becomes

$$\begin{cases} 1 - 3 + 2 = 0 \\ 2 + 3c + 2d = 0 \\ 1 - c + 2d = 0 \\ 6 = 6 \\ 4 + c^2 + 2d^2 = 6 \end{cases} \implies \begin{cases} 3c + 2d = -2 \\ -3c + 2d = -2 \\ c^2 + 2d^2 = 2 \end{cases} \implies \begin{cases} d = -1 \\ c = 0 \\ 0^2 + 2(-1)^2 = 2 \end{cases}$$

	$\chi(E)$	$\chi(1 \to 2)$	$\chi(1 \to 2 \to 3)$
1	1	1	1
$1'$	1	−1	1
2	1	0	−1

Consider now a representation of S_3:

$$U(e) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, U(1 \to 2) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, U(1 \to 3) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, U(2 \to 3) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$U(1 \to 2 \to 3) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, U(1 \to 3 \to 2) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Since all these matrices are unitary and the representation is reducible, then it is completely reducible. The character vector of U is

$$\vec{\chi}_U = \begin{pmatrix} 1 \cdot 3 \\ \sqrt{3} \cdot 1 \\ 0 \end{pmatrix}$$

For the previous results,

$$n_1 = \frac{1}{6}\sum_{i=1}^{3} |c_i|\chi_1^*(c_i)\chi_U(c_i) = \frac{1}{6}\vec{\chi}_1^\dagger \cdot \vec{\chi}_U = [...] = 1$$
$$n_{1'} = \frac{1}{6}\vec{\chi}_{1'}^\dagger \cdot \vec{\chi}_U = [...] = 0$$
$$n_2 = \frac{1}{6}\vec{\chi}_2^\dagger \cdot \vec{\chi}_U = [...] = 1$$

So this means that the representation is reducible to

$$U = 1 \oplus 2$$

Def. A *left-regular representation* of a finite group G is a representation which acts on G as a left translation. More precisely, consider the vector space $V = \mathbb{C}^{|G|}$ with base $\{e_g\}_{g\in G}$: the left translation is defined as

$$\begin{aligned} L :& G \to \mathrm{Perm}(G) \\ & g \to L_g \text{ s.t. } L_g(h) = g \cdot h \end{aligned}$$

and the regular representation is

$$\begin{aligned} & D(g) : V \to V \\ & e_h \to D(g)e_h = e_{g\cdot h} \\ & D\left(g\right) v = D(g)\left(\sum_{h\in G} v^h e_h\right) := \sum_{h\in G} v^h D(g) e_h = \sum_{h\in G} v^h e_{g\cdot h} \end{aligned}$$

In matrix form the element corresponding to $e_h, e_{h'}$ is

$$D(g)_{h,h'} = \begin{cases} 1 & \text{if } h = g \cdot h' \\ 0 & \text{otherwise} \end{cases}$$

this means that for $g = e$, the matrix is the identity of V; on the other hand, if $g \neq e$, the diagonal is always 0, which means all $D(g \neq e)$ are *traceless*:

$$\forall g \in G,\, \chi(g) = |G|\delta_{ge}.$$

Def. A *right-regular representation* of a finite group G is a representation which acts on G as a right translation:

$$R : G \to \mathrm{Perm}(G) \tag{13.1}$$
$$g \to R_g \text{ s.t. } R_g(h) = h \cdot g^{-1} \tag{13.2}$$

and the representation is similarly defined.

14 Tensors

Def. Given a vector space V of dimension n over the complex field, the *dual vector space* V^* is

$$V^* := \{f : V \to \mathbb{C} \text{ s.t. } f \text{ is linear}\}$$

V^* is a vector space with $\dim(V^*) = \dim(V)$.

Given a basis $\{e_i\} \subset V$, a natural choice of basis for V^* is the *dual basis*

$$\left\{e^{*i}\right\}, \text{ s.t. } e^{*i}(e_j) = \delta^i{}_j$$

If V is equipped with a scalar product $\langle \cdot, \cdot \rangle$ and the basis is orthonormal, this dual basis is precisely

$$e^{*i} := \langle e_i, \cdot \rangle \to e^{*i}(e_j) := \langle e_i, e_j \rangle = \delta^i{}_j$$

which is actually where the notations $e^{i*} = \langle e_i|$ and $e_j = |e_j\rangle$ come from.

Lesson 11, 04/11

Def. Given a vector space V, its dual V^* and their bases $\{e_i\} \subset V$, $\left\{e^{*i}\right\} \subset V^*$, the product between $f \in V^*$ and $v \in V$ is

$$\langle f, v \rangle := f(v)$$

and it is an inner product.

Def. Given V, W vector spaces and a linear function $f : V \to W$, the *pullback of* f, denoted as f^*, is the linear map

$$\begin{aligned} f^* : W^* &\to V^* \\ g &\to f^*(g) := g \circ f \end{aligned}$$

Since $g \in W^*$, $g : W \in \mathbb{C}$ is linear and so $f^* = g \circ f : V \in \mathbb{C}$ is a linear map, so it is an element of V^*.

Def. A *tensor* of type (p, q) over the vector space V is a function

$$T : \underbrace{V^* \times V^* ... V^*}_{p \text{ times}} \times \underbrace{V \times V ... V}_{q \text{ times}} \to \mathbb{C}$$

which is *linear in all its arguments*.

Note From the fact that $(V^*)^* = V$ follows that $(1, 0)$ tensors are vectors, while $(0, 1)$ tensors are dual vectors.

The components of a tensors are expressed in terms of the basis and dual basis of V, V^*:

$$T = T^{i_1 ... i_p}{}_{j_1 ... j_q} e_{i_1} ... e_{i_p} e^{j_1 *} ... e^{j_q *}$$

where the sum over all the $p + q$ indices is omitted for Einstein's notation.

Def. Given a (p, q) tensor R and a (l, m) tensor S, the *tensor product* of R and S, denoted as

$$T =: R \otimes S$$

is a tensor of type $(p + l, q + m)$ such that

$$S(f_1, ... f_{p+l}, v^1, ... v^{q+m}) = R(f_1 ... f_p, v^1, ... v^q) S(f_{p+1} ..., f_{p+l}, v^{q+1}, ... v^{q+m})$$

This is a bilinear, associative but not commutative operation.

Def. Given a (p, q) tensor, its *contraction of indices* is a $(p-1, q-1)$ tensor, denoted as

$$T_{c(i,j)}$$

where i is a contravariant index and j is a covariant index: it is defined so that

$$T_{c(i,j)}(f_1, ..., ...f_{p-1}, v^1, ...v^{q-1}) := \sum_{k=1}^{\dim V} T(\underbrace{f_1, ...f_{i-1}, e^{*k}, f_i, ...f_{p-1}}_{p \text{ arguments}}, \underbrace{v^1, ...v^{j-1}, e_k, ...v^{q-1}}_{q \text{ arguments}})$$

Note In coordinates this is

$$(T_{c(i,j)})^{\alpha_1,...\alpha_{p-1}}{}_{\beta_1,...\beta_{q-1}} = T^{\alpha_1,...\alpha_{i-1},k,\alpha_i,...\alpha_{p-1}}{}_{\beta_1,...,\beta_{j-1},k,\beta_j,...\beta_{q-1}}$$

where the sum over k is omitted.

Def. Given two vector spaces V, W of dimensions $\dim(V) = n$, $\dim(W) = m$, and with bases

$$\{e_i\} \subset V, \{e'_j\} \subset W$$

the *tensor product of* V, W is

$$V \otimes W = \left\{\Lambda^{ij} e_i \otimes e'_j, \text{ with } \Lambda^{ij} \in \mathbb{C}\right\}$$

and has dimension $\dim(V \otimes W) = \dim(V)\dim(W)$.
Note Not all elements of a product space can be expressed as a tensor product of vectors, i.e. **the existence of** v, w so that

$$\lambda_1 v_1 \otimes w_1 + \lambda_2 v_2 \otimes w_2 = v \otimes w$$

is **not** guaranteed.
This fact is at the base of the quantum mechanical concept of *entanglement*[9], for example a spinor state $|\Psi\rangle \in \mathbb{C}^2 \otimes \mathbb{C}^2$ such as

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|1\rangle \otimes |0\rangle + |0\rangle \otimes |1\rangle)$$

cannot be written as a linear combination in the form

$$|\Psi\rangle \neq (\alpha |0\rangle + \beta |1\rangle) \otimes (\gamma |0\rangle + \delta |1\rangle)$$

The concept of tensor product can also be generalized to be between dual spaces, between a vector and a dual space, and even between tensor product spaces with a similar definition: for example

$$V \otimes W^* = \left\{\Lambda^i{}_j e_i \otimes e'^{j*}\right\}$$

$$V \otimes V^* \otimes V = \left\{\Lambda^i{}_j{}^k e_i \otimes e^{*j} \otimes e_k\right\}$$

Note Consider a change of basis over the vector space V

$$e'_k = R^l_k e_l$$

and its dual base e^{*m} so that

$$e^{*m}(e_l) = \delta^m_l :$$

by requiring

$$e'^{*n}(e'_k) = \delta^n_k$$

[9] See the Bachelor's degree course "Quantum Mechanics II".

it is clear that

$$e'^{*n} = (R^{-1})^n_m e^{*m}$$

This also means that any vector does not change under change of basis:

$$v = v^i e_i = v^j \delta^i_j e_i = \underbrace{v^j (R^{-1})^l_j}_{=v'^l} \underbrace{R^i_l e_i}_{=e'_l} := v'^l e'_l$$

Def. Given two linear operators

$$A : V_1 \to V_2,\ B : W_1 \to W_2$$

their *tensor product* is the linear map

$$\begin{aligned} A \otimes B : V_1 \otimes W_1 &\to V_2 \otimes W_2 \\ \Lambda^{ij} e_i \otimes e'_j &\to (A \otimes B)(\Lambda^{ij} e_i \otimes e'_j) = \Lambda^{ij} A(e_i) \otimes B(e'_j) \end{aligned}$$

Note Consider $A \in \mathrm{Mat}(n \times n)$, $B \in \mathrm{Mat}(m \times m)$: their tensor product is a $nm \times nm$ matrix defined as

$$\begin{pmatrix} a_{11} & ... & a_{1n} \\ ... & ... & ... \\ a_{n1} & ... & a_{nn} \end{pmatrix} \otimes \begin{pmatrix} b_{11} & ... & b_{1m} \\ ... & ... & ... \\ b_{n1} & ... & b_{mm} \end{pmatrix} = \begin{pmatrix} a_{11}B & ...B & a_{1n}B \\ ...B & ...B & ...B \\ a_{n1}B & ...B & a_{nn}B \end{pmatrix}$$

this particular case of the tensor product of linear maps is called the *Kronecker product* and is general to rectangular matrices.
Note The trace is well behaved with respect to this operation, namely

$$\mathrm{Tr}(A \otimes B) = \mathrm{Tr}(A)\,\mathrm{Tr}(B).$$

Consider now the automorphism

$$L : V \otimes W \to V \otimes W$$

that can be represented as

$$L = L_{ijlk}(|i\rangle \otimes |j\rangle) \otimes (|l\rangle \otimes |k\rangle)$$

[things about trace]